

FD-11 Radiosonde Enclosure

0.6 mm thin-wall SLA housing for a long-duration LoRaWAN picoballoon payload

SOLE MECHANICAL ENGINEER · CAD, PROTOTYPING, ASSEMBLY, TESTING AND PRODUCTION RELEASE

The FD-11 is an ultralight radiosonde developed by Project Picoballoon for long-duration atmospheric observations from high-altitude weather balloons. Every gram of structure is a gram of lift margin, and lift margin sets the ceiling, so the housing is a two-part thin shell printed at **0.6 mm wall thickness** — thin enough that stiffness, printability and clip durability stop being independent problems and have to be solved as one.

6 g ENCLOSURE MASS — AGAINST AN 8 G BUDGET	0.6 mm NOMINAL WALL THICKNESS — 0.4 MM AT THE CLIPS	200 UNITS PRODUCED — EXTERNAL SLA VOLUME MANUFACTURE	85 × 23 × 18 ENVELOPE, MM
--	---	--	---

ARCHITECTURE

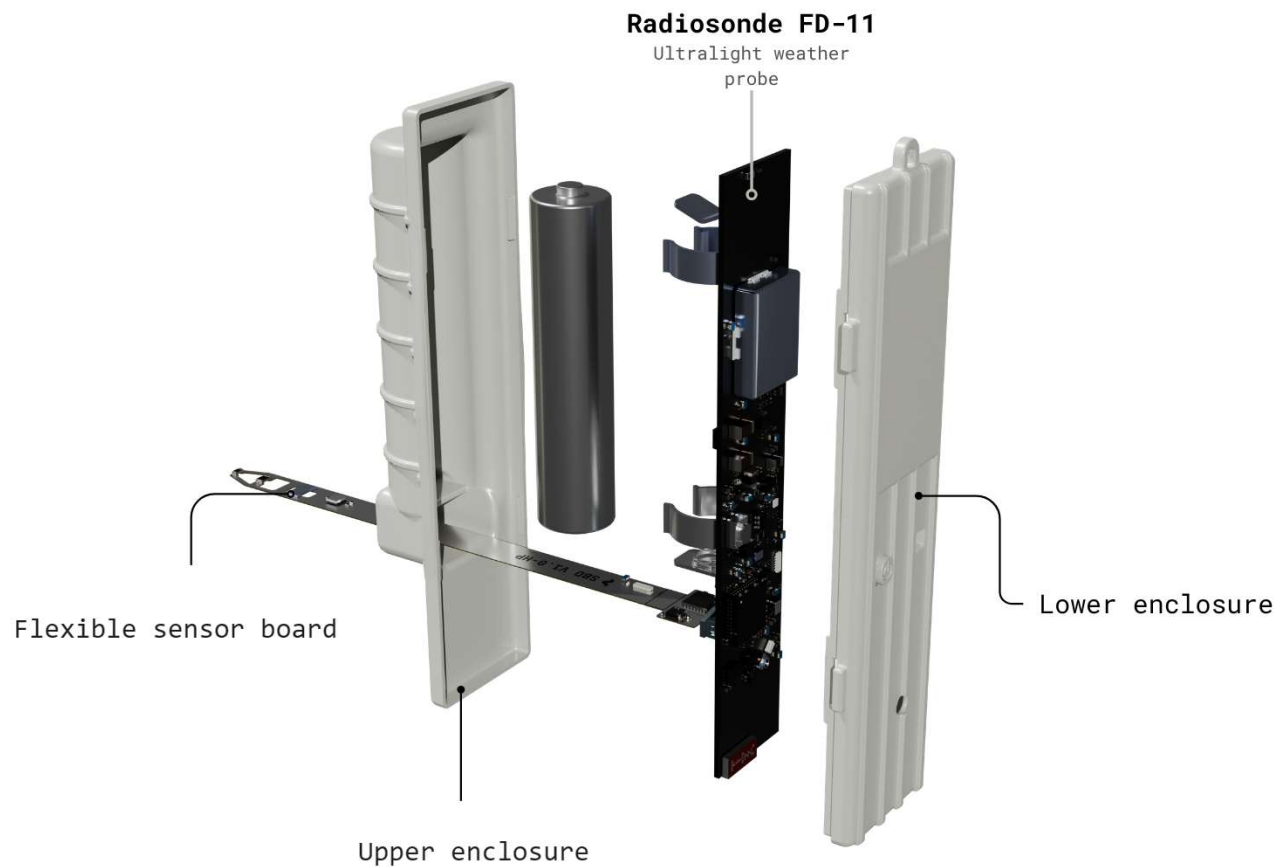


Fig. 1 – Exploded assembly. The shell closes on four internal clips; the AAA cell, the probe and the internal antenna sit inside the 85.3 × 22.6 × 18 mm envelope, and the flexible sensor board leaves the shell through a slot in the parting line.

MASS BUDGET

20 G FLIGHT ASSEMBLY

Enclosure · 6 g

30% OF FLIGHT MASS

Rest of flight assembly · 14 g

70% — LITHIUM AAA CELL, PROBE FD-11, TETHER

The cell was set by the thermal environment and the probe by the electronics design, so the mass budget had to be met by the structure — the only item the mechanical design controls. Individual cell and tether masses are inferred from the 20 g total rather than measured, so the remainder is shown as one segment TBD

Requirements and the clip problem

REQUIREMENT	RESULT
Mass ≤ 8 g — mission director	6 g
Contain probe, AAA cell and internal antenna — probe design, external	85.3 × 22.6 × 18 mm envelope
Carry the flight train: balloon attaches to the enclosure, probe hangs from it — system architecture	One separation, before the tether-ring fix; none since
Sensors in free air, thermally decoupled from the electronics — measurement requirement	Flexible sensor board exits the shell
Survive multi-day flight, target 50+ days — mission	Longest verified flight: ~4 days

No temperature requirement was formally specified, and there is no dedicated vent — the shell leaks through the clip gaps, the LED opening and the sensor-board slot, so equalization is a consequence of the architecture rather than an intent.

THE CLIP

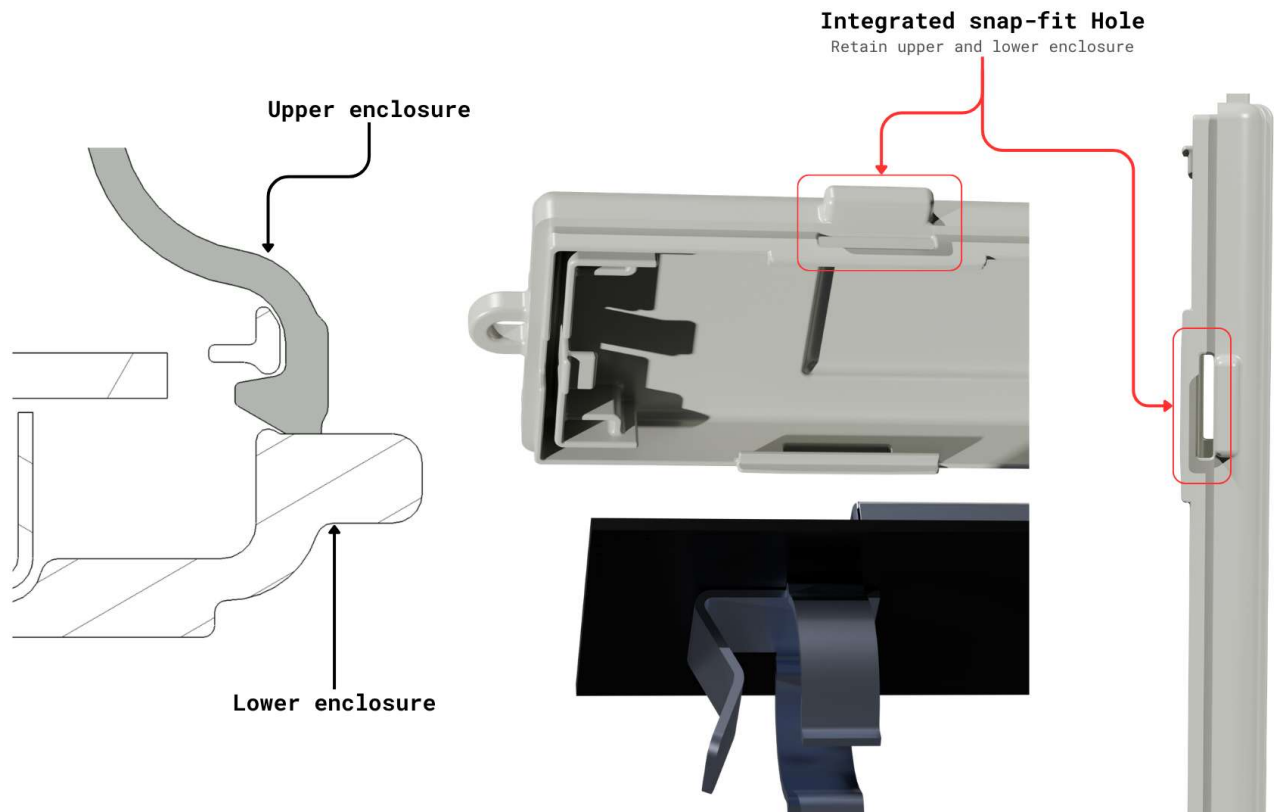


Fig. 2 – Section through the snap-fit. The clip is a 0.4 mm cantilever on the upper shell; it deflects inward as the shells close and returns into the receiving hole in the lower shell. Every opening in a 0.4 mm wall leaves a thin unsupported bridge above it – the ribs restore thickness locally without stiffening the surrounding wall, which would move the crack rather than prevent it.

Seven iterations

3–4 MONTHS

Three iterations were mine: clip geometry, snap-fit dimensions, and local ribs where openings left an unsupported bridge in the 0.4 mm wall. Three came from the probe changing underneath the enclosure — the power button, the sensor board and the LED each moved, and each moved an opening. The seventh followed the tether-ring failure in flight.

3 DESIGN · 3 INTERFACE · 1 FLIGHT FAILURE

Clips

Four clips close the shell. They tolerate roughly four or five open/close cycles; the flight condition is closed once and not reopened. No FEA was run. Low prototype cost, short turnaround and uncertainty in thin-wall printed-resin behaviour made empirical iteration the practical development approach.

Print orientation

Prototypes printed at 45°, chosen for printability: a smaller cross-section peels per layer, a closed thin-wall section cannot form a suction pocket, and resin drains. Orientation was fixed by the process, so clip behaviour had to be tuned through geometry.

What flight proved, and what it did not

FAILURE AND FIX

The closed ring where the tether attaches broke once, during flight. The enclosure is the structural link in the flight train, so a ring failure is a loss of the payload rather than a cosmetic problem: the probe separated from the balloon and fell. The ring thickness was increased, and there has been no recurrence on subsequent flights. **Ring thickness before and after the fix** TBD

BALLOON → TETHER RING → ENCLOSURE → PROBE

LOAD PATH – THE ENCLOSURE IS IN IT

VALIDATION

1.5 m drop

Onto concrete, fully assembled with cell and boards installed. Pass = shell stays closed and probe still transmitting. Impact 5.4 m/s, ≈0.29 J for the 20 g assembly. Ten units, five drops each.

–78 °C soak

Cold-soaked in dry ice, way below the coldest recorded flight temperature, then inspected for cracking. Soak duration: 30min

15 km

Maximum altitude flown. Minimum recorded temperature → -61C

53 flights

Across the program, all of them on the current frozen design. The enclosure carries the entire flight train, so every completed flight is considered a proof-load test of the shell and the ring.

The dry-ice soak was a static exposure with visual inspection. It does not cover mechanical loading at low temperature: SLA photopolymers become stiffer and less tough when cold, so clip engagement and impact resistance at flight temperature remain untested.

PRODUCTION

TWO BUILDS, 200 UNITS

Prototyping ran in-house in Formlabs Tough 2K, printed and assembled by hand. Production of 200 units went to JLC3DP in two builds. Internal test launches use CBY resin, sanded only, at €0.56 per set — the flights are not recovered, so cost per flight dominates. Units delivered to clients use 9600 resin with exterior matte white spray paint at €3.28 per set, where surface finish is part of the deliverable. The painted finish was specified for appearance, not as a UV barrier.

€0.56 / set

CBY RESIN, SANDED ·
INTERNAL TEST LAUNCHES ·
€112 AT 100

€3.28 / set

9600 RESIN, PAINTED ·
CLIENT DELIVERIES · €656
AT 100

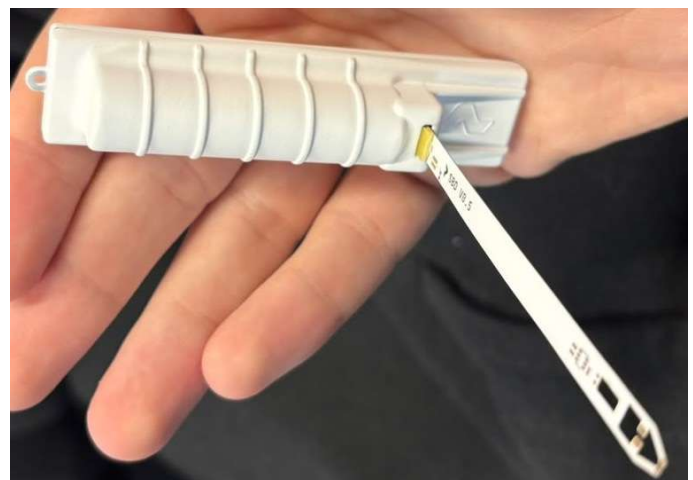


Fig. 3 – Client build: painted upper and lower shells closed on the probe.

The 45° orientation was qualified on the in-house prototypes, but orientation for the production run was chosen by the supplier, so the released parts were not necessarily built the way the prototypes were qualified. Flight heritage was also accumulated on the sanded CBY build, so it does not transfer directly to the delivered build.

WHAT I WOULD CHANGE

DESIGN FROZEN

Clip mechanism

The snap-fit is the weakest part of the design. It took the most iterations, caused the early assembly failures, and tolerates only a few open/close cycles. A more robust retention method is the first thing I would change.

50-day survival is unverified

No mission reached that duration. Flights ended when radio coverage was lost, not from hardware failure, so the target remains unproven. Longest verified exposure: **3days**.

UV is on the wrong build

The units that spend weeks in stratospheric UV are the unpainted sanded ones; the white coating sits on the client build, which lives indoors. Embrittlement of the 0.4 mm clips is unaddressed on exactly the parts that fly.